Multichannel Pulse-Coupled Neural Network-Based Hyperspectral Image Visualization

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***Abstract*— Hyperspectral Image (HSI) visualization, which aims at displaying as much material information of original images as possible on a trichromatic monitor with natural color, plays an important role in image interpretation and analysis. However, most of the HSI visualization methods only focus on presenting the detail information of a scene without providing natural colors and distinguishing land covers with similar colors. In order to address this problem, this article proposes a multichannel pulse-coupled neural network (MPCNN)-based HSI visualization method, which consists of the following steps. First, the MPCNN is proposed and explored to fuse the original HSI so as to obtain a fused band with rich spatial details. Then, a color mapping scheme is proposed to determine the weights of red, green, and blue (RGB) channels. Finally, the weighted RGB channels are stacked together for visualization. Experiments performed on four hyperspectral data sets demonstrate that the proposed method not only displays the HSI with nature colors but also improves the details in the image. The effectiveness of the proposed method is demonstrated in terms of both visual effect and objective indexes.**

***Index Terms*— Color mapping, hyperspectral image (HSI) visualization, multichannel pulse-coupled neural network (MPCNN), natural color display.**

I. Introduction

HYPERSPECTRAL satellite sensors are now able to provide hundreds of spectral bands, corresponding to a wide range of wavelengths. Thus, this type of image contains rich spatial–spectral information, which has been extensively applied in various aspects [1]–[9], including object recognition [10], [11], land-cover classification [12], [13], and anomaly detection [14], [15].

Due to the availability of a high number of spectral channels in hyperspectral data, the existing display devices that are designed to visualize images containing one or three channels cannot provide a quick overview of the hyperspectral image (HSI). Hence, in order to address this issue, an HSI visualization technique is developed, which aims at projecting hundreds of hyperspectral bands into red, green, and blue (RGB) channels. The major challenges behind this problem are two folds. The first is how to display as much information as possible from the original HSI. The second is how to show the natural colors of different land covers, and thus, the displayed image will be more suitable for human perception. In the past few decades, many HSI visualization approaches have been proposed [16]–[23], which can be generally classified into three categories: band selection-based methods, transform-based methods, and fusion-based methods.

The main idea of band selection methods is to directly select three representative bands as RGB channels according to some kind of rules for HSI visualization [24]–[29]. For example, some software tools, e.g., the environment for visualizing images (ENVIs) [30], provide an interactive way for users to manually select three bands for visualization. In addition, some advanced band selection methods have been developed, such as 1-bit transform (1BT) [16], orthogonal subspace projection-based method [26], and constrained energy minimization-based method [27]. Demir *et al.* [16] proposed a 1BT-based approach for low-complexity selection of suitable bands for the RGB color display of HSIs, in which the less-structured bands are first eliminated by using the 1BT of each band, and then, three suitable bands are selected from the remaining set of well-structured bands for visualization. Su *et al.* [24] adopted minimum estimated abundance covariance and linear prediction to select the three most informative bands for visualization. Furthermore, some advanced band selection methods were also developed. Yuan *et al.* [31] developed a dual-clustering-based band selection method with context analysis, in which a pairwise hyperspectral angle descriptor is designed by using the context information of a specific pixel. Yuan *et al.* [32] proposed a multitask sparsity pursuit (MSP) framework for displaying HSI. A random-projection-based descriptor is constructed using the sparse signal recovery power of compressive sensing. Then, a band

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search strategy is utilized to select the candidate bands followed by MSP-based criterion for assessing each possible band combination. Jacobson and Gupta [33] proposed a color matching function (CMF)-based method to create a consistent rendering, in which a fixed linear spectral weighting envelope was used to produce natural-looking images.

Different from band selection-based methods, the main idea of transform-based methods is to transform the original image into a new feature space by using a spectral transformation algorithm and select the first three features as RGB channels for HSI visualization. Among the transformation-based approaches, the independent component analysis (ICA) [20] and the principal component analysis (PCA) [34] are the most commonly used ones. For instance, in [21], a hue, saturation, and value (HSV) color space-based method is proposed, in which the PCA algorithm is first performed on the original image, and the first three features are mapped into HSV space for visualization. In [35], a segmented PCA (SPCA)-based method is proposed, in which the original image is first partitioned into three groups of subsets, and then, the first principal component of each subset is combined together for color display. Furthermore, some nonlinear transform methods were also used to display HSIs, such as faithful stochastic proximity embedding [36] and locality preserving projection [37].

In addition, some fusion-based approaches have also been developed for the display of HSIs [38]–[41], such as bilateral filtering-based method [40] and decolorization-based method [41]. Kotwal and Chaudhuri [40] proposed a bilateral filter (BF)-based visualization method, in which the weight of each band is determined by the BF. Kang *et al.* [41] developed a decolorization-based hyperspectral visualization (DHV) framework, which can well preserve the separability of different classes and details. However, these approaches mentioned above aim at retaining more information from the original image while not producing a resulting image with natural colors. Therefore, it is difficult to interpret land covers in hyperspectral scene. For instance, user may be confused when the tree is displayed in blue or the water is displayed in yellow. In order to overcome this issue, in [37], a manifold alignment (MA)-based framework was proposed to display HSI with natural colors by fusing an HSI and a high-resolution true-color image that was obtained from Google Earth. However, the high-resolution true-color image is required and registered accurately with the HSI, which is difficult in real applications. To improve this method, Liao *et al.* [42] proposed a constrained manifold learning (CML) method for HSI visualization, in which the spectral and spatial information are incorporated into the manifold structure by composite kernel. These methods both can produce a natural-looking resulting image. However, the obtained results are very similar to the RGB bands, which do not provide other important information, such as more spatial details and discrimination belonging to different land covers, for the following interpretation and analysis (see Fig. 1). In addition, the manifold learning-based methods are usually very time-computing because of the computation of the adjacency matrix in the high spectral dimension.

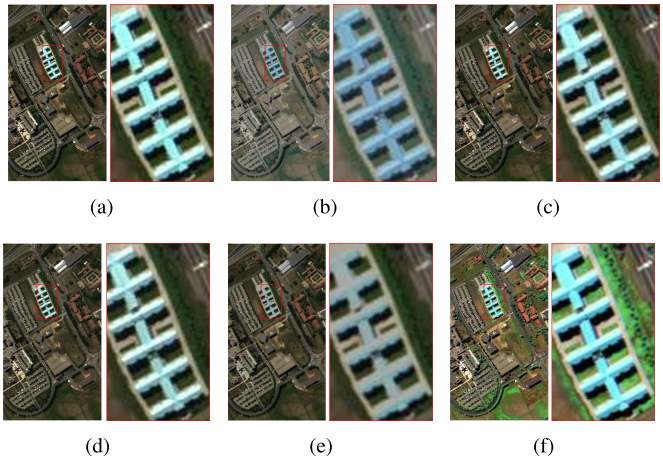

Fig. 1. Comparison of visualization results from RGB bands and other advanced methods. (a) RGB bands. (b) CMF method [33]. (c) MA method [37]. (d) CML method (feature-level learning) [42]. (e) CML method (instance-level learning) [42]. (f) Our method.

Inspired from recent development in a pulse-coupled neural network (PCNN) [43], known as the third-generation artificial neural network, it has been widely used in image processing, such as image segmentation [44], image fusion [45], shadow removal [46], and object detection [47], which also provides substantial potential for HSI visualization. In order to provide more information for image interpretation and analysis, in this article, we propose an adaptive multichannel PCNN (MPCNN) method for HSI visualization since it closely resembles the human visual system and does not require any training process that reduces the computing time compared with other artificial neural networks. The main difference between our method and other state-of-the-art methods is that our method not only preserves the spatial details and the true colors of land covers but also improves the discrimination belonging to different land covers in comparison with other methods. The proposed method mainly consists of the following steps: first, to make full use of the interdependencies among different spectral bands, the MPCNN model is constructed and applied to fuse the original HSI to obtain a fused band with rich details. Then, the fused band is mapped into RGB channels via a designed color mapping scheme. Finally, the obtained RGB bands are stacked together for visualization. Experiments demonstrate that the proposed method not only can yield natural color visualization results but also can greatly improve visualization performance in discrimination of land covers. The main contributions and advantages of this article can be concluded as follows.

1) A novel MPCNN-based hyperspectral visualization method is proposed. The proposed method not only preserves the details of original HSI but also provides the natural colors of land covers.
2) In order to overcome the drawbacks of the traditional PCNN, an MPCNN is constructed for the first time and applied for the visualization of HSIs.
3) Compared to the state-of-the-art visualization methods, our method provides superior performance in distinguishing land covers with similar colors such as trees and meadows.

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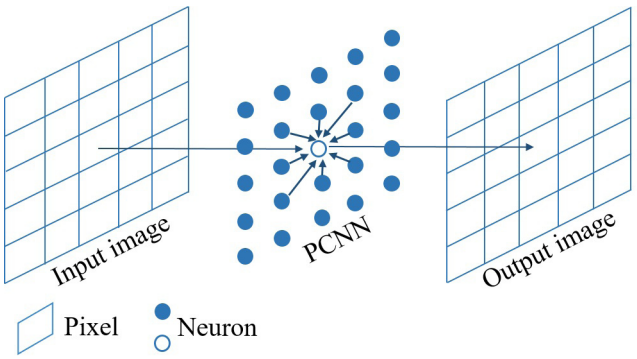

Fig. 2. Connection model of PCNN neuron.

The following part of this article is organized as follows. The PCNN is reviewed in Section II. The proposed MPCNN-based visualization method is described in Section III. Experimental results are given in Section IV. Finally, conclusions are presented in Section V.

II. PCNN

PCNN is a biologically inspired neural network model [48], which is obtained by studying the synchronous neuronal burst phenomena in the cat visual cortex. Johnson and Padgett [49] first illustrated the effectiveness of the PCNN model in image processing, such as image fusion, object detection, and image enhancement. Different from the most existing artificial neural networks, PCNN is a 2-D single-layer connection network and does not require any training process, in which image pixels and network neurons exist with a one-to-one correspondence (see Fig. 2). Each neuron in the PCNN model contains three parts: the receptive domain, the linking modulation, and the pulse generator. A simplified PCNN model can be described as follows.

The receptive domain

$$\mathbf{F}_{ij}(n) = \mathbf{S}_{ij}(n) \tag{1}$$

$$\mathbf{L}_{ij}(n) = V_L \sum_{kl} \mathbf{W}_{ij,kl}\mathbf{Y}_{ij}(n-1) \tag{2}$$

where n is the number of iterations, and the indexes i and j denote the pixel location in each band. V_L denotes the amplitude of the linking input $\mathbf{L}$. $\mathbf{F}_{ij}(n)$ is the feeding input that receives the external stimulus $\mathbf{S}$, and $\mathbf{L}_{ij}(n)$ is the linking input that connects with the neighboring neurons by constant synaptic weight

$$\mathbf{W} = \begin{bmatrix} 0.5 & 1 & 0.5 \\ 1 & 0 & 1 \\ 0.5 & 1 & 0.5 \end{bmatrix}.$$

The role of the receptive domain is to receive the input signals including external inputs and surrounding neuron stimuli.

The linking modulation

$$\mathbf{U}_{ij}(n) = e^{-\alpha_L}\mathbf{U}_{ij}(n-1) + \mathbf{F}_{ij}(n)(1+\beta\mathbf{L}_{ij}(n)) \tag{3}$$

where $\mathbf{U}_{ij}(n)$ is the internal state of the neuron. β indicates the linking parameter. In the linking modulation, two inputs are modulated by a linking strength β to generate an internal activity $\mathbf{U}_{ij}(n)$.

The pulse generator

$$\mathbf{Y}_{ij}(n) = \begin{cases} 1, & \text{if } \mathbf{U}_{ij}(n) > \mathbf{T}_{ij}(n-1) \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

$$\mathbf{T}_{ij}(n) = e^{-\alpha_T}\mathbf{T}_{ij}(n-1) + V_T\mathbf{Y}_{ij}(n) \tag{5}$$

where V_T is the amplitude of the dynamic threshold $\mathbf{T}$. When $\mathbf{U}_{ij}(n)$ is greater than the dynamic threshold $\mathbf{T}_{ij}(n)$, current neuron is fired $\mathbf{Y}_{ij}(n) = 1$, and then, the dynamic threshold will increase by amplitude V_T. Otherwise, if the neuron is not fired, the dynamic threshold will decay by a factor $e^{-\alpha_T}$.

In this article, PCNN is adopted to merge all bands of HSI in the process of visualization because of the following reasons: first, PCNN is a model based on iterative calculation and does not require any training process, which is able to tackle the high-dimensional HSI more efficiently compared to the existing artificial neural networks. Second, PCNN developed by Echoron's cortical model has been demonstrated the effectiveness in image fusion.

III. Proposed Method

Fig. 3 shows the proposed HSI visualization method, which mainly consists of the following steps: construction of MPCNN, fusion of spectral bands, and color mapping.

A. Construction of MPCNN

The traditional PCNN model faces two key challenges. One is that the PCNN model is designed to tackle the visible image while HSI usually consists of hundreds of bands. Thus, when the PCNN model is used to deal with each channel of an HSI, the correlations among different channels are ignored. Second, the conventional PCNN model has many parameters such as linking strength, decay coefficients, and various amplitudes. Moreover, the values of these parameters are given manually, which significantly affects the operational applicability of such networks in real applications.

To overcome these drawbacks, in this article, the MPCNN is constructed to deal with HSIs, in which the parameters are determined by an automatic design method. The MPCNN model can be described as follows.

The receptive domain

$$\mathbf{F}_{ij}^{k}(n) = \mathbf{S}_{ij}^{k}(n) \tag{6}$$

$$\mathbf{L}_{ij}^{k}(n) = exp(-\alpha_L)\mathbf{L}_{ij}^{k}(n-1) + f^{k}(\mathbf{Y}(n-1)) \tag{7}$$

where n is the number of iterations. $\mathbf{F}_{ij}^{k}(k = 1, 2, .., K)$ denotes the feedback input at the (i, j) position of the kth channel. $\mathbf{L}_{ij}^{k}$ is the linking input at the (i, j) position of the kth channel. α_L is the attenuation time constant. $f^{k}(\cdot)$ is the feed function, which refers to the influence of surrounding neurons on the current neuron. Here, $f^{k}(\mathbf{Y}(n-1)) = \sum_{a,b}\boldsymbol{\omega}_{ij,ab}\mathbf{Y}_{ij}(n-1)$, where a and b denote the dislocation in a symmetric neighborhood around one pixel, and

$$\boldsymbol{\omega} = \begin{bmatrix} 0.707 & 1 & 0.707 \\ 1 & 0 & 1 \\ 0.707 & 1 & 0.707 \end{bmatrix}.$$

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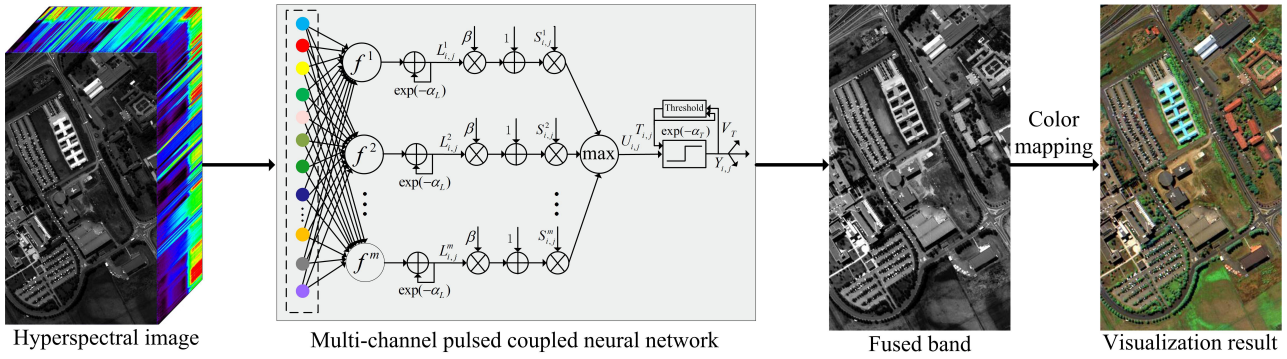

Fig. 3. Schematic of the proposed MPCNN-based visualization method.

Compared with the traditional PCNN, the receptive domain of the proposed model is able to receive multiple external stimuli and output of surrounding neurons at the same time, which is more suitable for dealing with high-dimensional HSI.

The modulation domain

$$\mathbf{U}_{ij}(n) = \max\left\{\mathbf{F}_{ij}^{k}(n)\left(1+\beta_k \mathbf{L}_{ij}^{k}(n)\right)\right\} \tag{8}$$

where $\mathbf{U}_{ij}$ is the total internal activity. β_k is the linking strength of kth channel, which indicates the importance of the kth channel. This step makes full use of the interdependencies among different spectral bands against traditional PCNN, which can merge well the finer details of different bands.

The pulse-generated domain

$$\mathbf{Y}_{ij}(n) = \begin{cases} 1, & \mathbf{U}_{ij}(n) > \mathbf{T}_{ij}(n-1) \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

$$\mathbf{T}_{ij}(n) = e^{-\alpha_T}\mathbf{T}_{ij}(n-1) + V_T \mathbf{Y}_{ij}(n) \tag{10}$$

where $\mathbf{Y}_{ij}$ is the output pulse of the neuron after internal processing, in which $\mathbf{Y}_{ij}$ outputs a pulse in iteration n ($\mathbf{Y}_{ij}(n) = 1$) only when $\mathbf{U}_{ij}$ is greater than $\mathbf{T}_{ij}$. $\mathbf{T}_{ij}$ is the dynamic threshold. α_T and V_T denote the attenuation time constant and the normalized constant, respectively.

In the proposed MPCNN model, there are four free parameters requiring setting together, and it is very hard to set all parameters to be the optimal values. In this situation, the parameters usually are set manually based on experience or repeated experiments. Inspired by the automatic parameter setting method for image segmentation [50], in this article, the automatic parameter setting method in [50] is introduced into the MPCNN, which makes it easier in real applications. Specifically, four free parameters in the MPCNN model, i.e., α_L, β_k, α_T, and V_T can be calculated as follows:

$$\alpha_L = ln\left(\frac{1}{\sigma(\mathbf{S})}\right) \tag{11}$$

$$\beta = \frac{(S_{\max}/S') - 1}{6} \tag{12}$$

$$V_T = e^{-\alpha_L} + 1 + 6\beta \tag{13}$$

$$\alpha_T = \ln\left(\frac{V_T/S'}{\frac{1-e^{-3\alpha_L}}{1-e^{-\alpha_L}} + 6\beta e^{-\alpha_L}}\right) \tag{14}$$

where $\sigma(\mathbf{S})$ denotes the standard deviation (SD) of the input image $\mathbf{S}$. S' and S_{max} indicate the normalized Ostu threshold and the maximum intensity of the input image, respectively. More details about the parameter settings can be found in [50].

B. Fusion of Spectral Bands

To make the visualized image have rich details, the details in different bands should be fully utilized. In this article, the MPCNN is used to fuse the HSI so as to obtain a fused band. Specifically, the main steps of the MPCNN-based fusion can be concluded as follows.

1) The original HSI $\mathbf{I}$ is normalized between 0 and 1 as follows:
$$\bar{\mathbf{I}}_{ij} = (\mathbf{I}_{ij} - \min(\mathbf{I}_{ij}))/(\max(\mathbf{I}_{ij}) - \min(\mathbf{I}_{ij})). \tag{15}$$
2) The parameters of the MPCNN model are initialized, i.e., $\mathbf{L}_{ij}(0) = \mathbf{U}_{ij}(0) = \mathbf{Y}_{ij}(0) = 0$, $\mathbf{T}_{ij}(0) = 1$. All neurons are not fired.
3) The values of the parameters used in MPCNN model, i.e., α_L, β_k, α_T, and V_T, are computed automatically according to (11)–(14).
4) The normalized $\bar{\mathbf{I}}$ is used as the feeding input to motivate the MPCNN neurons, and then, $\mathbf{L}_{ij}(n)$, $\mathbf{U}_{ij}(n)$, $\mathbf{Y}_{ij}(n)$, and $\mathbf{T}_{ij}(n)$ are calculated according to (6)–(10).
5) Equations (6)–(10) are iterated until all neurons are activated. The resulting image can be selected as follows:
$$\mathbf{F}_{ij} = \mathbf{S}_{ij}^{k}, \quad \text{if } \mathbf{U}_{ij} = \mathbf{U}_{ij}^{k} \tag{16}$$
where $\mathbf{U}_{ij}^{k} = \mathbf{S}_{ij}^{k}(1+\beta_k \mathbf{L}_{ij}^{k})$.

C. Color Mapping

In order to obtain the visualization result with natural colors, the fused band is mapped into a color image. The proposed color mapping technique is based on an assumption that the spectral range of HSI is available. In real applications, the spectral range of hyperspectral sensor usually is known, and thus, these bands, which are falling in red (0.63−0.69 μm), green (0.525−0.605 μm), and blue (0.45−0.515 μm) wavelength ranges, can be easily obtained. Specifically, the RGB bands located in RGB wavelength ranges are first selected and fused together

$$\tilde{\mathbf{V}}_t = \sum_{t\in\{R,G,B\}} \mathbf{I}_t / N_t \tag{17}$$

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Algorithm 1 MPCNN for HSI Visualization

Input:
Input hyperspectral image $\mathbf{I}$;
Output:
Visualization result $\mathbf{V}$;
1: **Begin**
2: The normalized hyperspectral image $\bar{\mathbf{I}}$ is computed using (15).
3: Initialize parameters. $\mathbf{L} = \mathbf{U} = \mathbf{Y} = 0$, $\mathbf{T} = 1$.
4: According to (11)-(14), compute the values of parameters, i.e., α_L, β_k, α_T, and V_T.
5: **For** $n = 1 : N$ **Do**
$\mathbf{S}^k = \bar{\mathbf{I}}^k$
$\mathbf{L}^k = e^{-\alpha_L}\mathbf{L}^k + \mathbf{K} \bigotimes \mathbf{Y}^k$
$\mathbf{U} = max\{\mathbf{S}^k(1 + \beta_k \mathbf{L}^k)\}$
If $\mathbf{U}_{ij} > \mathbf{T}_{ij}$, then $\mathbf{Y}_{ij} = 1$
else $\mathbf{Y}_{ij} = 0$
$\mathbf{T} = e^{\alpha_T}\mathbf{T} + V_T\mathbf{Y}$
End For
6: If $\mathbf{U}_{ij} = \mathbf{S}^k_{ij}(1 + \beta_k \mathbf{L}^k_{ij})$ then
7: $\mathbf{F}_{ij} \leftarrow \mathbf{S}^k_{ij}$
8: end If
9: According to (18), compute the weight of each channel.
10: According to (19), compute the final RGB bands for visualization.
11: Construct the visualization result $\mathbf{V} = \{\mathbf{R}, \mathbf{G}, \mathbf{B}\}$.
12: **Return V**
13: **End**

where t denotes the band index of the HSI spectrum. N_t is the total number of bands falling into the R, G, and B wavelength ranges, respectively.

Then, the weights of RGB channels in the visualization result are calculated as follows:

$$\boldsymbol{\omega}_t = \frac{\tilde{\mathbf{V}}_t}{\sum_{t \in \{R,G,B\}} \tilde{\mathbf{V}}_t}. \tag{18}$$

Finally, the RGB bands for visualization are obtained by stacking the weighted RGB channels

$$\mathbf{V_t} = \boldsymbol{\omega}_t \times \mathbf{F} \tag{19}$$

where $\mathbf{V}_t$, $t \in \{R, G, B\}$ are the final RGB channels for HSI visualization.

The pseudocode of the proposed MPCNN method is presented in Algorithm 1. First, the MPCNN is proposed and used to fuse the original HSI so as to obtain a fused band with rich details. Then, the fused band is mapped into a natural-looking visualization image. It should be mentioned that one iteration of Algorithm 1 may be not able to achieve the optimal solution. In this situation, more iterations may be required in performing the MPCNN so as to obtain the better visualization performance. In order to obtain the satisfactory performance, the number of iterations N is set to be 30 based on the following experiment.

IV. Experiments

In this section, the experimental setup is first presented. Then, the influence of the parameter to the visualization performance is analyzed. Next, the effectiveness of different PCNN models is discussed. Finally, the visualization results are discussed.

A. Experimental Setup

1) Data Sets: In our experiments, four hyperspectral data sets, i.e., the University of Pavia image, the Cuprite image, the Houston image, and the "Cloud" image are used to test the effectiveness of different methods.

The University of Pavia image[1] was captured by the reflective optics system imaging spectrometer (ROSIS) sensor over the Engineering school of the University of Pavia, Italy. The image contains 610 × 340 pixels, with 103 spectral bands ranging from 0.43 to 0.86 μm and spatial resolution of 1.3 m/pixel. Its RGB image is constructed by the 46th, 31st, and 10th bands.

The Cuprite image[2] was captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor in Las Vegas, Nevada, U.S. It has 224 channels with 250 × 190 pixels, ranging from 0.4 to 2.5 μm. After removing the noisy channels (1–2 and 221–224) and water absorption channels (104–113 and 148–167), there remain 188 channels for visualization. The corresponding RGB image is generated by its 28th, 19th, and 10th bands.

The Houston image[3] was acquired by the Hyperspectral Image Analysis Laboratory and the National Center for Airborne Laser Mapping (NCALM) over the University of Houston campus and the neighboring urban area, which is released during the 2018 Data Fusion Contest of the IEEE Geoscience and Remote Sensing Society (GRSS). The size of this image is 4172×1202 pixels with a spatial resolution 1 m. It is composed of 48 bands ranging from 0.38 to 1.05 μm. The corresponding RGB image is generated by its 20th, 13th, and 6th bands.

The fourth data set[4] was captured by the AVIRIS sensor, named "Cloud" image. This data set contains 224 bands ranging from 0.4 to 2.5 μm and a spatial resolution of 20 m/pixel. The spatial dimension of the image is 951 × 611 pixels. The corresponding RGB image is constructed by its 28th, 19th, and 10th bands.

2) Compared Methods: To verify the effectiveness of the proposed method, several widely used HSI visualization techniques and RGB bands, i.e., dominant set extraction(DSE)-based band selection [51], MSP [32], BF [40], PCA [34], SPCA [35], DHV [41], MA [37], CML [42], and RGB bands are adopted to be compared with the proposed method. The default parameters setting presented in the corresponding publications are utilized for all experiments. Furthermore, in order to make the proposed method be reproducible, the code in this article will be shared online[5].

[1] http://lesun.weebly.com/hyperspectral-data-set.html
[2] http://lesun.weebly.com/hyperspectral-data-set.html
[3] http://hyperspectral.ee.uh.edu/?page_id=1075
[4] http://www.escience.cn/people/duanpuhong/sj.html
[5] https://github.com/PuhongDuan

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a) DSE: The DSE method is a dual-clustering-based band selection method. A pairwise hyperspectral angle feature of HSI is extracted. Then, a dual-clustering framework is proposed by using the extracted features. Finally, a groupwise strategy is used for the representation of clusters, and the selected three bands are taken as the visualization results. The corresponding code is available on the author's homepage[6].

b) MSP: The MSP method is a band selection-based visualization method by MSP. The band selection is modeled as a multitask sparse learning problem, in which the immune clonal strategy is used to search the candidate combinations of the hyperspectral band.

c) BF: The BF method is a hierarchical framework-based visualization technique, in which the weight for each band is calculated with the BF. The fusion process utilizes all band information at all locations in all bands, unlike other approaches where all bands are assigned the same weight.

d) PCA: The original HSI is transformed into feature space, and then, the first three principal components are chosen as the RGB bands for visualization.

e) SPCA: The spectral bands of original HSI are partitioned into three subsets, and then, the first principal component of each subset is combined together for visualization.

f) DHV: The DHV method is an image decolorization-based visualization framework, in which the image decolorization technique is used to map three fused color images into RGB channels. The code of this method is available on the Github[7].

g) MA: The MA method is a natural color visualization technique, in which the HSI and the high-resolution color image are fused via MA. The code of this method is available on the Github[8].

h) CML: The CML method is a CML-based visualization technique, in which a composite kernel is applied in manifold learning to fuse both the spatial and spectral information in the embedded space. The code of this method is available on the Github[9].

3) Quality Indexes: To evaluate objectively visualization performance of different methods, six extensively used objective indexes, i.e., separability of features (SF) [37], SD [26], average gradient (AG) [32], entropy (EN) [32], fusion factor (FF) [52], and quantitative correlation analysis (QCA) [53] are adopted. For convenience, the six quality indexes are described briefly as follows.

1) SF measures how well distinct pixels are transferred to distinguishable colors. The core idea is to calculate the distance between two pixels in the color space. SF is defined as follows:

$$\mathrm{SF} = \frac{1}{(N-1)^2} \sum_{x_i \neq y_i} d(x_i, y_i) \tag{20}$$

where $d(x_i, y_i)$ denotes the Euclidean distance between a pair of pixels x_i and y_i, and N is the total number of pixels.

[6]https://gzhu.weebly.com/
[7]https://github.com/PuhongDuan/Hyperspectral-image-visualization
[8]https://github.com/sakurafan/MATransfer
[9]https://github.com/sakurafan/CML

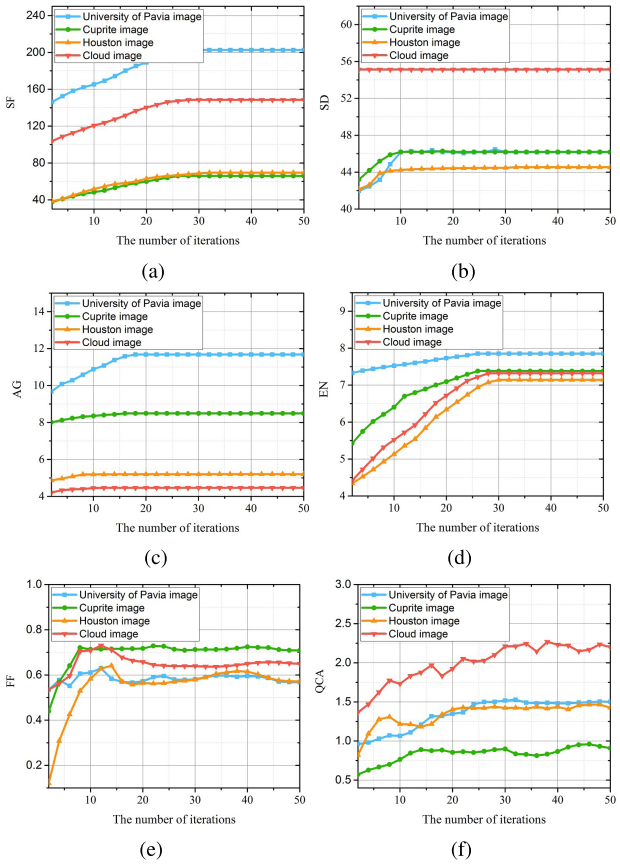

Fig. 4. Influence of the parameters, i.e., the number of iterations of MPCNN (N), to the visualization performance of the proposed method. (a) SF. (b) SD. (c) AG. (d) EN. (e) FF. (f) QCA.

2) SD is the contrast degree of the visualization result as follows:

$$\mathrm{SD} = \left(\frac{1}{N-1} \sum_{i=1}^{N} (x_i - \bar{x})^2 \right)^{\frac{1}{2}} \tag{21}$$

where $\bar{x}$ denotes the mean value of the pixels, and x_i denotes the value of the ith pixel.

3) AG measures the clarity of the visualization result as follows:

$$\mathrm{AG} = \frac{1}{N} \sum_{x} \sum_{y} \left(\sqrt{I_x^2 + I_y^2} \right) \tag{22}$$

where I_x and I_y indicate the difference in the x-axis and y-axis directions, respectively.

4) EN is the degree of information preservation in the visualization result

$$\mathrm{EN} = -\sum_{i} p_i \ln p_i \tag{23}$$

where p_i represents the probability density.

5) FF assesses the similarity between the original HSI and the visualization result

$$\mathrm{FF} = \frac{\sum_{i=1}^{B} MI(I_i, I_F) H(I_i)}{\sum_{i=1}^{B} H(I_i)} \tag{24}$$

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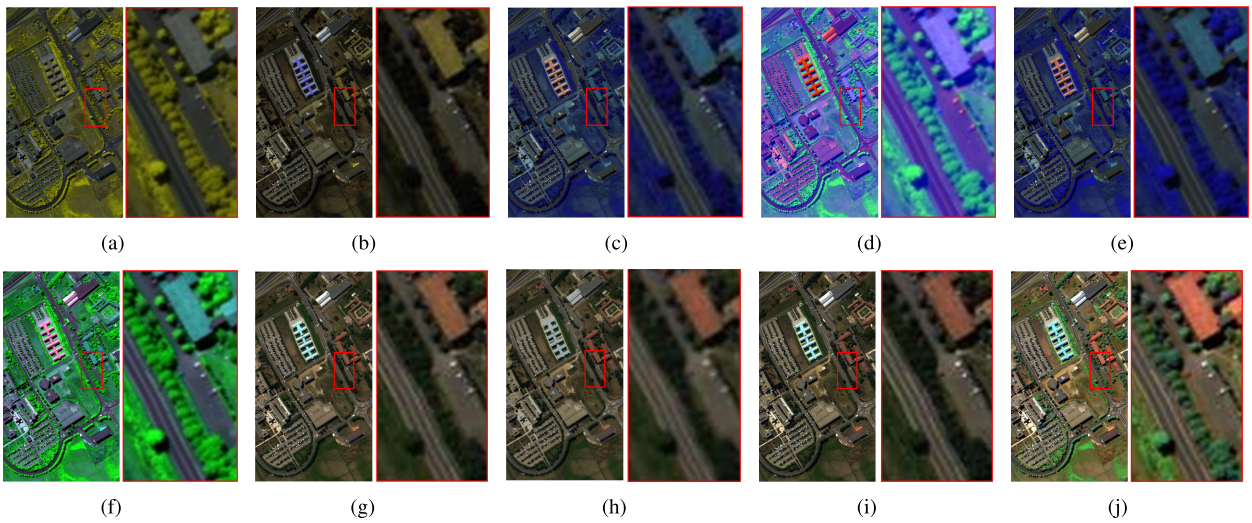

Fig. 5. Visualization results of different methods on the University of Pavia data set. (a) DSE method [51]. (b) MSP method [32]. (c) BF method [40]. (d) PCA method [34]. (e) SPCA method [35]. (f) DHV method [41]. (g) MA method [37]. (h) CML method [42]. (i) RGB bands. (j) Proposed MPCNN method.

where $MI(I_i, I_F)$ implies the amount of mutual information between the input image and the visualization result. $H(I_i)$ is the entropy of the ith image.

6) QCA means that how much information in original HSI is injected into the visualization result

$$\begin{aligned} \text{QCA} &= R(I, I_F) = R(I_F \cap (I_1 \cup I_2 \cup \cdots \cup I_B)) \\ &= R\left(\bigcup_{i=1}^{B} (I_F \cap I_i)\right) \end{aligned} \tag{25}$$

where $R(I_F, I_i)$ denotes the correlation information entropy between I_F and I_i.

B. Parameter Analysis

In this section, the influence of the parameter, i.e., the number of iterations (N) of MPCNN, to the visualization performance of the proposed method is discussed on four hyperspectral data sets. Fig. 4 presents the influence of the parameter to the visualization performance. It can be observed in Fig. 4 that when N is relatively small, the visualization performance is unsatisfactory. Furthermore, the visualization performance of the proposed method tends to be stable when N is larger than 30. Therefore, in this article, the parameter N is set as $N = 30$.

C. Analysis the Effectiveness of the MPCNN

In order to demonstrate the effectiveness of the proposed MPCNN model, the influence of different PCNN models to the visualization performance of the proposed framework is analyzed. This experiment is performed on the University of Pavia data set. Several representative PCNN models, i.e., classic PCNN (PCNN) [48], PCNN with automatic parameter setting (APCNN) [50], and MPCNN without automatic parameter setting (WMPCNN), are adopted to be compared with the proposed MPCNN model. Table I presents the visualization performance of the proposed framework with different PCNN models. It can be seen that the APCNN model can obtain better visualization performance since the automatic parameter setting method yields better parameters for visualization. Furthermore, the proposed MPCNN model with automatic parameter setting obtains the best visualization performance. The main reason is that the MPCNN can fully exploit the correlations among spectral bands over other PCNN models.

TABLE I
VISUALIZATION PERFORMANCE OF THE PROPOSED FRAMEWORK WITH DIFFERENT PCNN MODELS

Methods	SF	SD	AG	EN	FF	QCA
PCNN [48]	164.60	41.15	10.14	7.13	0.43	1.27
APCNN [50]	171.90	43.06	10.99	7.33	0.46	1.32
WMPCNN	187.87	45.07	11.20	7.63	0.54	1.41
MPCNN	**202.58**	**46.16**	**11.68**	**7.85**	**0.58**	**1.51**

D. Visualization Results

The experiment is first performed on the University of Pavia image. Fig. 5 shows the visualization results of all considered methods. As shown in this figure, for the DSE method, the differences belonging to various land covers are quite small. The MSP method makes a slight improvement in terms of the separability belonging to different objects. However, some land covers such as the meadows and roads in the resulting image are not able to be displayed clearly [see the closeup in Fig. 5(b)]. For the BF method, the color differences of various objects tend to increase since the spatial information in the original image is fused together using the hierarchical scheme. However, some ground objects such as roofs and roads do not have a clear outline [see the closeup in Fig. 5(c)]. For the PCA method, the resulting image has a hazy appearance, and the background color in the visualization result becomes "violet" [see Fig. 5(d)]. The resulting image obtained by the SPCA method is similar to the one obtained by the BF method [see Fig. 5(e)]. The DHV method improves

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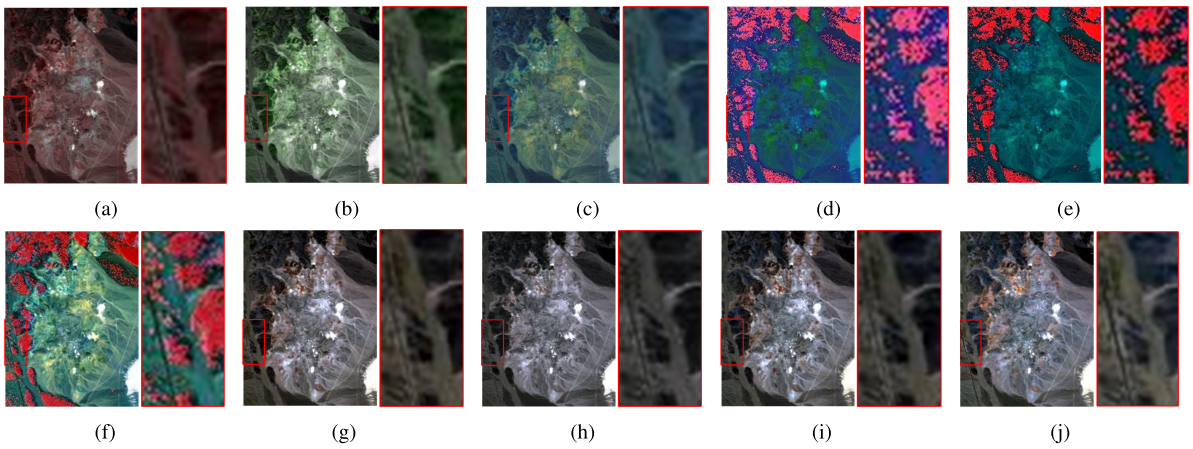

Fig. 6. Visualization results of different methods on the Cuprite data set. (a) DSE method [51]. (b) MSP method [32]. (c) BF method [40]. (d) PCA method [34]. (e) SPCA method [35]. (f) DHV method [41]. (g) MA method [37]. (h) CML method [42]. (i) RGB bands. (j) Proposed MPCNN method.

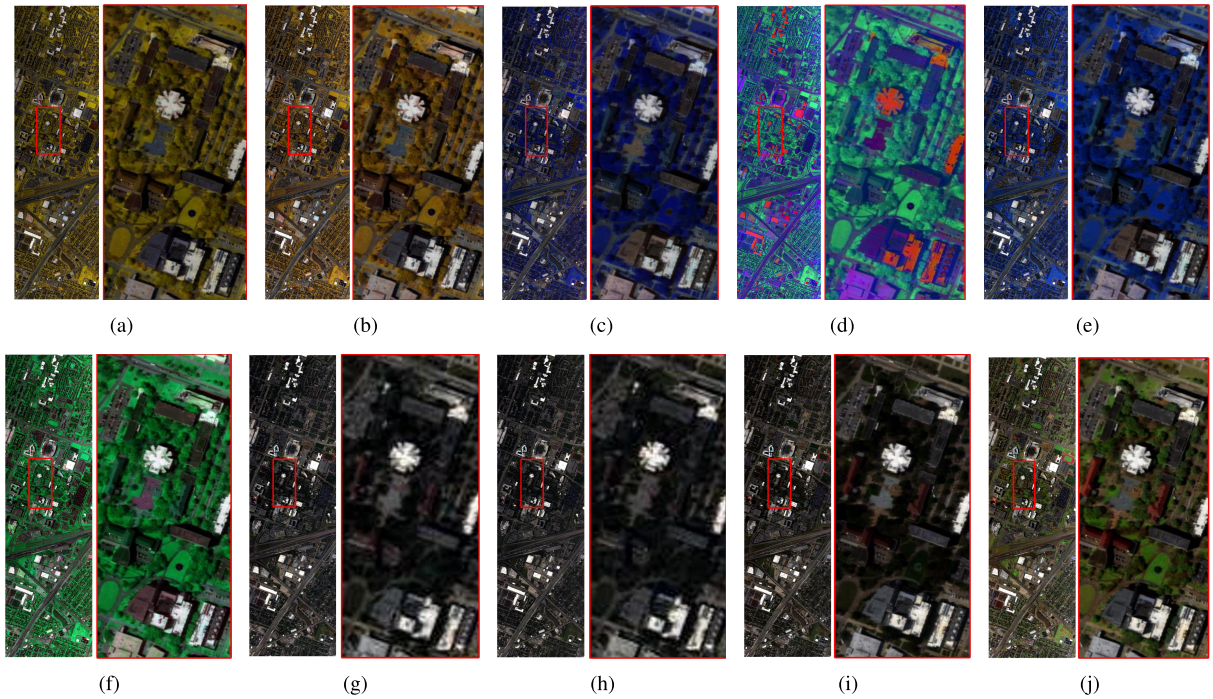

Fig. 7. Visualization results of different methods on the Houston data set. (a) DSE method [51]. (b) MSP method [32]. (c) BF method [40]. (d) PCA method [34]. (e) SPCA method [35]. (f) DHV method [41]. (g) MA method [37]. (h) CML method [42]. (i) RGB bands. (j) Proposed MPCNN method.

the color differences belonging to different land covers, but the background color is inconsistent with the real scene [see the closeup in Fig. 5(f)]. The RGB image can provide the natural color resulting image. However, some land covers such as trees and meadows in the resulting image become dark because of the limitation of the hyperspectral sensor in the spatial resolution. Although the MA and CML methods can obtain the resulting images with natural colors and preserve the spectral information of HSI, the resulting images are very similar to the RGB image. By contrast, it can be seen that the proposed method not only can produce a natural-looking resulting image but also can improve the color differences belonging to similar land covers, e.g., trees and meadows, which enables users to understand the scene information much easier.

The second experiment is conducted on the Cuprite data set. The resulting images obtained by different methods are shown in Fig. 6. A closeup view of each image is shown in the right side. In this example, For the DSE and BF method, the resulting images are not satisfactory since the finer details cannot be displayed clearly. The MSP method improves the whole contrast of the resulting image. However, the color differences of different objects are still small. In the images produced by the PCA, SPCA, and DHV methods, there exist noiselike spots. Different from the previous visualization results, the RGB image has natural colors for different objects. However, some finer details cannot be observed clearly [see the closeup in Fig. 6(i)]. The MA and CML methods provide the natural colors, but they are not able to preserve the spatial

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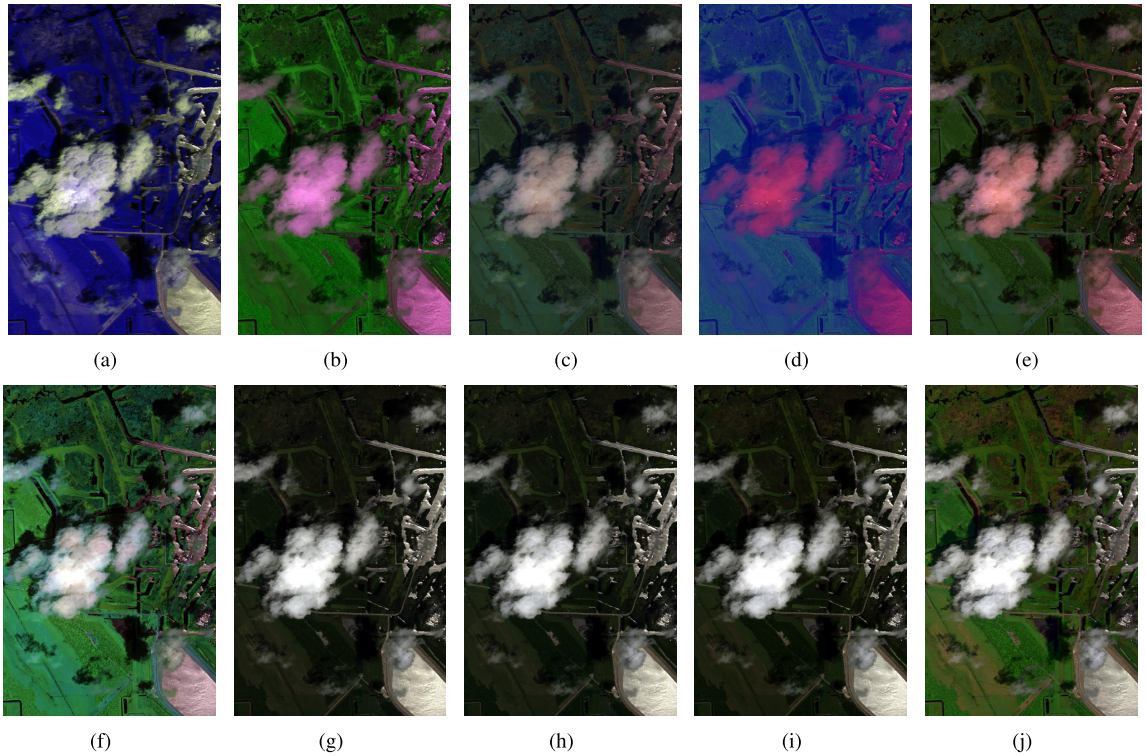

Fig. 8. Visualization results of different methods on the Cloud data set. (a) DSE method [51]. (b) MSP method [32]. (c) BF method [40]. (d) PCA method [34]. (e) SPCA method [35]. (f) DHV method [41]. (g) MA method [37]. (h) CML method [42]. (i) RGB bands. (j) Proposed MPCNN method.

TABLE II

OBJECTIVE EVALUATION OF THE DSE [51], MSP [32], BF [40], HSV [20], PCA [34], SPCA [35], DHV [41], MA [37], CML [42], RGB BANDS, AND OUR METHOD FOR THE UNIVERSITY OF PAVIA AND CUPRITE DATA SETS

Methods	University of Pavia data set						Cuprite data set					
	SF	SD	AG	EN	FF	QCA	SF	SD	AG	EN	FF	QCA
DSE	137.21	27.03	5.14	6.38	0.46	1.01	48.64	34.03	5.89	7.06	0.67	0.54
MSP	116.51	22.59	4.79	6.28	0.60	0.48	50.20	28.53	4.97	6.88	0.65	0.88
BF	121.50	22.88	5.33	6.61	0.57	0.94	43.19	27.66	4.54	6.67	0.68	0.75
PCA	162.81	43.37	11.01	7.72	0.62	0.91	40.12	40.55	6.08	6.35	0.72	0.65
SPCA	126.52	20.92	5.19	6.62	0.53	0.94	46.52	41.37	6.29	6.18	0.73	0.68
DHV	195.40	43.88	11.20	**7.86**	0.55	1.34	57.04	42.45	7.81	7.09	0.73	0.73
MA	113.40	44.05	9.59	7.27	0.61	0.77	63.65	43.92	7.52	7.10	0.60	0.84
CML	109.06	45.48	9.51	7.32	0.62	0.79	63.25	44.01	7.96	7.23	0.71	0.85
RGB bands	136.45	42.57	10.18	7.42	0.57	0.75	64.15	44.52	8.27	7.25	0.58	0.82
Our method	**202.58**	**46.16**	**11.68**	7.85	**0.58**	**1.51**	**65.91**	**46.19**	**8.49**	**7.38**	**0.76**	**0.89**

details of HSI [see Fig. 6(g) and (h)]. In Fig. 6(h), it can be observed that the proposed method not only can obtain a very natural tone but can also preserve the spatial details of the original HSI.

The third experiment is performed on the Houston data set. Fig. 7 shows the different visualization images obtained by the studied methods. As shown in this figure, the DSE and MSP method do not perform very well in distinguishing different ground objects. The BF and SPCA methods also produce the same phenomenon in the resulting images. The PCA method improves the discrimination of different land covers. However, some spatial details are still not visible in the resulting image. For the DHV method, the color differences of some land covers such as houses and trees are relatively small. The MA and CML methods cannot preserve the natural colors of RGB image in this example. The RGB image provides a low contrast, and some objects such as trees and grasses are also not distinguishable effectively. By contrast, the proposed method can obtain a trichromatic image with natural colors while preserving the rich spatial details.

The fourth experiment is conducted on the "Cloud" image. Fig. 8 presents the visualization results obtained by different methods. It can be seen that the proposed method still obtains a satisfactory result, which is in accordance with the previous experiments performed on the other images. Specifically, the proposed method not only obtains satisfactory performance on displaying HSIs with natural colors and preserving rich detail information but also improving the discrimination of different land covers.

The objective metrics of the different resulting images are presented in Tables II and III. In Tables II and III, it can be observed that the proposed method obtains competitive

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TABLE III

OBJECTIVE EVALUATION OF THE DSE [51], MSP [32], BF [40], HSV [20], PCA [34], SPCA [35], DHV [41], MA [37], CML [42], RGB BANDS, AND OUR METHOD FOR THE HOUSTON AND CLOUD DATA SETS

Methods	Houston data set						Cloud data set					
	SF	SD	AG	EN	FF	QCA	SF	SD	AG	EN	FF	QCA
DSE	45.01	40.25	4.72	5.67	0.34	1.07	132.57	48.20	3.86	6.28	0.42	2.07
MSP	47.12	43.99	4.92	5.58	0.50	1.18	135.16	52.19	4.03	6.67	0.48	1.92
BF	49.25	42.48	4.94	6.31	0.49	0.99	138.53	46.79	3.12	6.63	0.63	2.05
PCA	64.16	42.25	5.10	6.59	0.39	1.16	117.29	37.10	2.32	6.17	0.59	1.96
SPCA	49.42	40.24	4.81	6.25	0.40	1.31	134.92	46.24	3.04	6.82	0.59	1.92
DHV	62.58	45.04	5.13	7.10	0.53	1.25	144.02	52.75	4.20	7.03	0.57	2.18
MA	67.64	43.01	3.20	6.87	0.51	0.62	138.73	52.60	3.49	6.76	0.49	1.71
CML	66.72	42.03	2.97	6.74	0.49	0.59	130.92	50.02	3.52	6.79	0.48	1.59
RGB bands	68.34	42.62	4.43	6.83	0.44	0.65	139.41	51.22	2.77	6.55	0.46	1.72
Our method	**70.18**	**45.87**	**5.29**	**7.21**	**0.54**	**1.42**	**148.45**	**55.12**	**4.46**	**7.32**	**0.67**	**2.20**

TABLE IV

COMPUTING TIME OF THE DSE [51], MSP [32], BF [40], PCA [34], SPCA [35], DHV [41], MA [37], CML [42], AND OUR METHOD FOR THE UNIVERSITY OF PAVIA, CUPRITE, HOUSTON, AND CLOUD DATA SETS

Data sets	Computing time of all methods (in seconds)								
	DSE	MSP	BF	PCA	SPCA	DHV	MA	CML	Our method
University of Pavia	4.14	43.23	343.85	**0.66**	1.49	2.77	9833.83	10818.61	15.85
Cuprite	2.41	35.26	295.17	**0.61**	0.86	0.71	946.15	998.84	6.56
Houston	29.69	168.59	1475.29	**4.89**	14.82	36.9	196859.67	284911.28	198.45
Cloud	22.46	124.02	845.25	**2.74**	7.16	16.25	62923.04	82493.15	103.8

TABLE V

CLASSIFICATION PERFORMANCE OF THE SVM CLASSIFIER PERFORMED ON THE RESULTING IMAGES OBTAINED BY THE DSE [51], MSP [32], BF [40], PCA [34], SPCA [35], DHV [41], MA [37], CML [42], RGB BANDS, AND THE PROPOSED METHOD, RESPECTIVELY, ON THE UNIVERSITY OF PAVIA DATA SET WITH TEN TRAINING SAMPLES PER CLASS. NUMBERS IN THE PARENTHESIS INDICATE THE STANDARD VARIANCES OF THE ACCURACIES OBTAINED IN TEN REPEATED EXPERIMENTS

Class	Classification accuracies obtained by different methods (in %)									
	DSE	MSP	BF	PCA	SPCA	DHV	MA	CML	RGB bands	Our method
1	**95.91(2.98)**	83.79(6.43)	90.25(4.31)	92.38(2.96)	90.96(4.66)	91.51(4.36)	75.66(13.8)	73.74(11.7)	78.58(12.6)	77.15(2.48)
2	**83.06(2.47)**	78.27(3.09)	81.93(4.22)	80.95(2.93)	80.78(3.82)	80.77(3.77)	80.11(4.60)	79.03(4.90)	79.79(3.64)	81.05(2.40)
3	30.51(3.41)	30.22(3.78)	35.92(3.45)	**40.12(4.45)**	39.33(5.73)	38.08(5.58)	25.97(2.77)	27.23(3.17)	27.41(3.71)	29.71(2.65)
4	50.17(5.76)	40.82(8.01)	55.59(13.6)	51.10(6.40)	51.05(7.07)	49.65(8.02)	51.98(11.1)	45.12(7.87)	56.18(15.1)	**70.33(13.1)**
5	64.35(13.9)	97.10(3.71)	94.81(5.45)	93.66(7.44)	94.58(5.29)	91.76(5.46)	92.80(4.88)	95.57(3.94)	96.31(2.10)	**97.36(2.84)**
6	37.21(4.92)	39.6(25.23)	37.06(10.3)	37.36(8.84)	37.03(9.92)	34.27(7.93)	35.86(14.7)	25.42(7.14)	41.75(26.0)	**57.80(19.4)**
7	35.44(8.42)	38.71(7.89)	43.38(5.36)	42.83(6.68)	**44.27(7.82)**	44.03(8.14)	32.31(8.34)	32.71(7.05)	31.24(4.46)	37.27(4.63)
8	63.29(5.51)	54.97(14.4)	67.29(7.49)	**72.53(10.2)**	68.09(8.80)	72.47(6.16)	61.50(5.48)	57.12(13.1)	58.83(11.9)	64.18(6.90)
9	**99.94(0.07)**	72.63(12.7)	99.53(0.35)	99.91(0.13)	99.79(0.20)	99.73(0.27)	91.81(5.05)	94.40(3.75)	89.90(4.84)	98.81(1.45)
OA	62.41(1.52)	56.56(7.51)	64.74(5.25)	64.90(4.65)	64.33(4.23)	63.44(3.13)	60.89(4.18)	58.93(1.53)	60.30(5.95)	**70.15(1.89)**
AA	62.21(2.77)	59.57(3.05)	67.31(2.65)	67.87(2.44)	67.32(2.28)	66.92(2.15)	60.05(7.37)	54.37(5.36)	62.22(3.43)	**68.19(1.97)**
Kappa	53.15(1.79)	46.20(6.79)	55.92(5.38)	56.09(4.63)	55.42(4.26)	54.43(3.47)	49.75(7.40)	44.15(4.96)	50.00(6.09)	**60.66(1.83)**

performances in terms of SF, SD, AG, EN, FF, and QCA values, which further demonstrates the effectiveness of the proposed method.

Finally, in order to evaluate the computational efficiency of the eight approaches on the four hyperspectral data sets, the running time (in seconds) of different methods was recorded using MATLAB on a Laptop with 2.6 GHz and 8-GB RAM. Table IV shows the running time of different methods. It can be seen that the computing time of the proposed MPCNN method is moderate with respect to other studied methods (takes about 15.85 s to process an image with the size of 610×340 pixels). The reason is that the MPCNN used in the proposed method takes an iterative way to achieve the optimal performance. Designing a GPU framework for hyperspectral visualization is also an interesting topic.

E. Classification Results

It is difficult to quantitatively evaluate the quality of the visualization result since the reference image (ground truth) does not exist in practice. In order to further demonstrate the advantage of the proposed method, a supervised classifier, i.e., support vector machine (SVM) [54], is adopted to assess the performance of different visualization results

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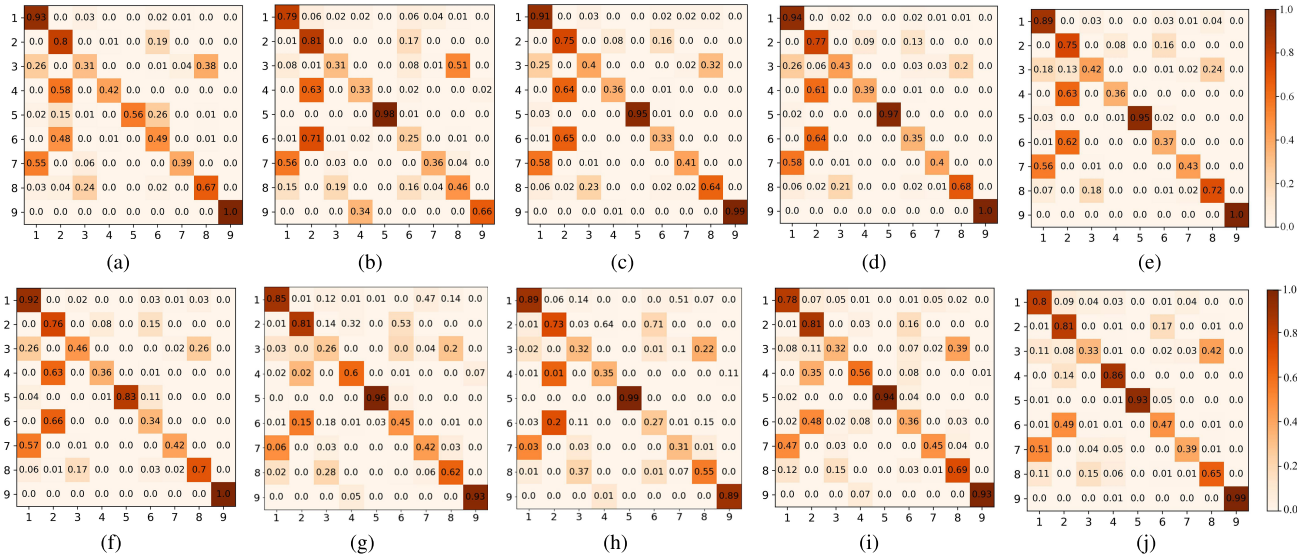

Fig. 9. Confusion matrices obtained by different methods for the University of Pavia data set. Numbers on the coordinate axes indicate different classes. (a) DSE method [51]. (b) MSP method [32]. (c) BF method [40]. (d) PCA method [34]. (e) SPCA method [35]. (f) DHV method [41]. (g) MA method [37]. (h) CML method [42]. (i) RGB bands. (j) Our method.

TABLE VI
NUMBERS OF TRAINING AND TEST SAMPLES

Class	Train	Test
Asphalt	10	6621
Meadows	10	18639
Gravel	10	2089
Trees	10	3054
Metal sheets	10	1335
Bare soil	10	5019
Bitumen	10	1320
Bricks	10	3672
Shadows	10	937
Total	90	42686

on the University of Pavia data set. It should be mentioned that this experiment is only tested on the University of Pavia data set since there is no standard ground truth for the other three data sets. The SVM Classifier is implemented using the LIBSVM library with a fivefold cross-validation technique. If the resulting image achieves higher classification accuracy, it means that the separabilities of different land covers in the visualization result are more obvious. Three widely used objective metrics, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient, are adopted to evaluate the classification performance.

The classification performance for the University of Pavia image is presented in Table V. From the ground truth, ten samples per class are randomly chosen as training samples [see Table VI]. It can be seen in Table V that the classification accuracies obtained by our method are the highest among all studied methods in terms of OA, AA, and Kappa coefficient, which demonstrates that the proposed method can greatly improve the separabilities belonging to different land covers. Furthermore, the individual accuracy of trees shows obvious improvement against other compared methods. This phenomenon also demonstrates that the proposed method can indeed make the trees easier to identify, which is consistent with the visual effect. In addition, confusion matrices of different methods for the Pavia University data set are shown in Fig. 9. It can be observed that the proposed method shows superior performance in distinguishing the land covers with similar colors, e.g., trees and meadows. However, for other studied methods, most of the fourth class (trees) are classified into the second class (Meadows). In contrast, our method can effectively discriminate between trees and meadows.

V. CONCLUSION

In this article, we propose a novel MPCNN model for HSI visualization, which is composed of two parts, i.e., the MPCNN-based fusion and color mapping. Specifically, we propose a newly designed MPCNN model to fuse the bands of an HSI so as to preserve the detailed information of the original HSI data. Furthermore, a simple yet effective color mapping scheme is proposed to overcome the false color display problem caused by traditional methods. Experimental results also demonstrate that the proposed MPCNN method can significantly improve the visualization performance in terms of detail preservation and natural color visualization compared to other studied approaches. As a possible future work, we are planning to design a novel natural-color rendering method without any prior knowledge.

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